

Alp Efe Karalar

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EDUCATION

The Pennsylvania State University | *B.S. in Computer Engineering* | **University Park, PA**

Dec 2024

Relevant Courses: OS Design, Comm. Networks, Computer Architecture, Data Structures & Algorithms, Neural Networks

Senior Capstone: Advanced Vehicle Team, Perception Department - Developed real-time computer vision models for autonomous vehicle navigation, achieving 3rd place overall in SAE AutoDrive Challenge II Competition

TECHNICAL SKILLS

Languages: Python, C/C++, JavaScript, TypeScript, Java, SQL, Bash, R, MATLAB

Technologies: Linux (Arch, Ubuntu, Debian), PyTorch, AWS, Docker, PostgreSQL, Git, Nginx, systemd

Concepts: Networking (TCP/IP, HTTP/HTTPS, DNS), Neural Networks, CI/CD, REST APIs, embedded systems

EXPERIENCE

The Pennsylvania State University | *Research Assistant* | **University Park, PA**

Dec 2025 - Present

- Build Python-based data matching pipelines to link >10,000 firms across a billion job postings using entity resolution techniques
- Process job posting datasets to extract hiring patterns, temporal trends, and employer-position relationships for research analysis
- Deploy fuzzy matching algorithms to resolve entity duplicates and ensure accurate company identification across large datasets
- Scale data infrastructure to handle decades of longitudinal employment data for labor market analysis in the finance sector

DementiAnalytics | *Software Engineer* | **University Park, PA**

Feb 2025 - Present

- Construct end-to-end ML-powered platform for dementia diagnosis, building React/Expo mobile application and Flask backend API
- Establish REST API integrating speech analysis models with patient data, processing audio biomarkers across 5-stage ML pipeline
- Implement HIPAA-compliant data handling architecture and authentication system for sensitive medical research data
- Design multiple web applications for research data visualization, model performance monitoring, and public-facing project websites

Bakkal Co. | *Software Engineer Intern* | **Pleasanton, CA**

Jun 2023 - Nov 2023

- Engineered comprehensive testing infrastructure for React Native mobile application supporting iOS and Android platforms
- Identified and documented critical bugs during testing cycles, working with senior engineers to implement fixes before release
- Integrated test suite with CI/CD pipeline, enabling automated test execution on pull requests

The Pennsylvania State University | *Learning Assistant (CMPEN270: Digital Design)* | **University Park, PA**

Jan 2022 - May 2024

- Troubleshoot technical issues for 250+ students per semester, creating documentation and video tutorials to improve student success
- Managed course operations, delivered lectures, and created course content while leading a team of learning assistants

PROJECTS

[Rust Load Balancer & Reverse Proxy](#) | Rust, Tokio, HTTP/1.1, TLS/SSL, Docker, Systems Programming

- Built production-ready layer 7 load balancer in Rust with Tokio async runtime, handling 5,000+ concurrent connections with <20MB memory overhead
- Implemented 4 load balancing algorithms with health checking and automatic failover
- Achieved <5ms p99 latency under 10,000 req/s load with TLS termination and connection pooling, providing 60% latency reduction

[Custom LISP Interpreter](#) | C, Parser Combinators, REPL, Language Design/Implementation

- Built a fully functional Lisp interpreter featuring parser, evaluator, REPL, and standard library
- Implemented core language features: S-expressions, lambdas, closures, error handling, file I/O, and string support
- Developed custom grammar using parser combinators (mpc library) to parse and evaluate expressions recursively

[Linux RAID Storage System \(mdadm\)](#) | C, Memory Management, LRU Cache, Systems Programming

- Developed Linux mdadm-inspired RAID emulator integrating 16 disks into a virtual block device with mount, read, write operations
- Implemented write-through LRU caching mechanism, achieving 87.9% cache hit rate and 30% reduction in average read latency

[LKML Dashboard - AI-Powered Kernel Development Tool](#) | Python, Flask, React, TypeScript, SQLite

- Engineered platform processing 600+ daily Linux Kernel Mailing List emails through automated parsing, thread reconstruction, and AI summaries

- Designed Flask REST API with 10 endpoints and SQLite FTS5 full-text search, achieving sub-second performance

[Embedded Linux Audio Processor](#) | Linux, RT Kernel, Armbian, JACK Audio, Guitarix, systemd

- Architected a real-time audio processing system on Orange Pi Zero using RT Kernel Armbian and JACK Audio
- Implemented automated boot sequences with systemd for production-ready guitar amplification without manual configuration